SECTION A

QUESTION 1 (compulsory)

(a) Explain various parameters used in classification and identification of plant diseases. **(12 marks)**

(b) Explain how Fluorescence In-Situ Hybridization (FISH) works as a molecular method of disease identification. Further state the advantages of this molecular method. **(8 marks)**

SECTION B

QUESTION 2

(a) In details, discuss plant disease tetrahedron **(8 marks)**

(b) Define the following terms
i. Stridulation
ii. Phytopathogenesis
iii. Biotrophs
iv. Phytoanticipins **(8 marks)**

(b) Describe various distinct stages involved in pathogenesis **(4 marks)**

QUESTION 3

(a) Insects are living organisms that belong to the class *insecta* and phylum *athropoda*. State and describe five (5) "*Orders*" into which insects can be classified including examples of insects that belong to such *Orders*. **(10 marks)**

(b) Briefly explain how bacteria reproduce **(5 marks)**

(c) Describe the differences between viruses and viroids **(5 marks)**

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PLANT PROTECTION (NR430) 2023 , TEST:1 TIME: 1hr: 30 min

Answer all the questions.

1. What do you understand by the following terms? Similar

(a). Hemibiotroph (b). Disease syndrome (c).Necrotroph (d). Predisposition (8 marks)

2. State and explain different parameters used to identify and describe plant diseases (12 marks)

3. Describe various symptoms associated with necrosis in fungal disease infestation in plants (10 marks)